D06 · Cold Chain & Specialized Logistics

04 · Packaging & unitization for transportation

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Transport packaging and unitization design the protective package, the case, the pallet pattern and the load so that product survives the distribution environment while using as much of the vehicle's cube as possible.

2 WHY IT MATTERS

Packaging sits on a trade-off: **too little protection raises damage and claims; too much wastes cube and freight**. Testing against recognized protocols such as ASTM D4169 or ISTA gives objective evidence of performance before field damage does. Pallet patterns and stack heights are often left untouched for years, which makes unitization one of the cheapest freight savings available.

3 HOW IT IS DONE

1. **Measure damage and cube by product family.** Damage rate, claim cost and units per trailer establish the baseline.
2. **Test before you change.** Run ASTM D4169 or ISTA sequences on current and proposed designs.
3. **Optimize the pallet pattern and load.** Interlocking, overhang and stack height for stability and fill.
4. **Decide on total cost.** Material + freight + damage + handling, not material cost alone.

IN PRACTICE

Test before you scale

Field damage is a very expensive way to discover that packaging is inadequate for the distribution environment.

Evaluate packaging changes on total cost, not material cost alone

A cheaper box that increases damage or reduces cube utilization is usually more expensive overall.

Optimize unitization as aggressively as primary packaging

Pallet patterns and stack heights are frequently left unchanged for years even as product dimensions evolve.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case — Consumer durables manufacturer** — Damage rate on a key product family was 2.8% with frequent “crushed corner” claims.

Initiative	Impact on Damage	Impact on Cube / Cost
Reinforced corner posts + better pallet interlocking	Damage -41%	Neutral on cube
Optimized pallet pattern (same protection)	Neutral	+6% units per truck
Elimination of unnecessary void fill	Neutral	-3% packaging material cost

> **Result:** Combined annual benefit approximately $1.4M from reduced damage claims plus freight and material savings. Payback on packaging engineering work was under four months.

HOW IT IS MEASURED

Damage Rate % (transport-related)

(Units or shipments with transport damage / Total units or shipments) × 100

Benchmark: Depends on product; track trend and cost of damage, not only frequency

Cube Utilization % / Units per Trailer

Actual product volume or units / Available trailer volume or theoretical maximum

Benchmark: Higher is better provided damage rate does not increase

WHAT YOU WOULD USE

Packaging & Load Optimization

Cape Pack / TOPS / specialized packaging design tools — Pallet pattern and cube optimization software.

TMS load-building modules — 3D load optimization that respects packaging and stacking constraints.

4 SO WHAT

Engineer packaging on total cost. Baseline damage and cube, test to a standard, optimize unitization and never judge a box by its price alone.

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